

Week 6 — LLN and CLT, Duality Theory, Conic Programming

AD7031 Optimization Under Uncertainty • Instructor: Prof. H. Park • Oct 3, 2026

1. Two classical results in probability theory

$\tilde{\xi}_1, \tilde{\xi}_2, \tilde{\xi}_3, \dots$: an infinite sequence of **independent and identically distributed (i.i.d.)** random variables,

each with expected value μ and variance σ^2

1) Law of large numbers (strong law, SLLN)

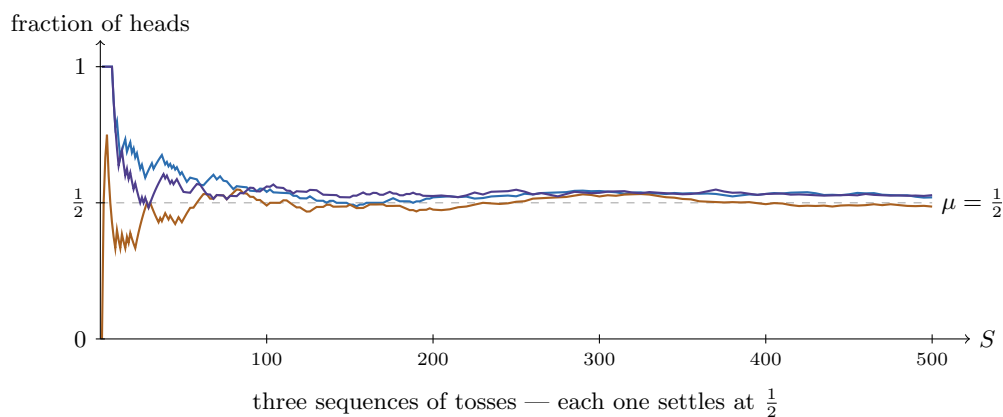
$$\mathbb{P}\left(\lim_{S \rightarrow \infty} \frac{1}{S} \sum_{i \in [S]} \tilde{\xi}_i = \mu\right) = 1$$

sample average $\longrightarrow$ expected value

Example toss a fair coin: $\tilde{\xi}_i = \begin{cases} 1 & \text{if toss } i \text{ is head} \\ 0 & \text{if toss } i \text{ is tail} \end{cases}$

$$\mu = \frac{1}{2}, \quad \sigma^2 = \mathbb{E}[\tilde{\xi}_i^2] - \mu^2 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

$$\frac{1}{S} \sum_{i \in [S]} \tilde{\xi}_i = \text{fraction of heads in the first } S \text{ tosses}$$



2) Central limit theorem (CLT)

$$\tilde{\chi}_S = \sum_{i \in [S]} \tilde{\xi}_i \quad \mathbb{E}[\tilde{\chi}_S] = S\mu, \quad \text{Var}(\tilde{\chi}_S) = S\sigma^2$$

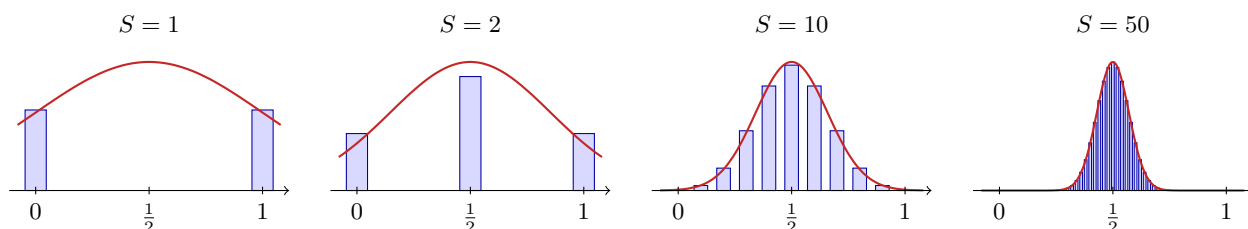
for large S , the random variable $\frac{\tilde{\chi}_S - S\mu}{\sigma\sqrt{S}}$ is approximately **standard normal**:

$$\mathbb{P}\left(\frac{\tilde{\chi}_S - S\mu}{\sigma\sqrt{S}} \leq r\right) \longrightarrow \mathbb{P}(\tilde{\zeta} \leq r) \quad \text{as } S \rightarrow \infty \quad \forall r \in \mathbb{R}, \quad \tilde{\zeta} \sim \mathcal{N}(0, 1)$$

whatever the distribution of $\tilde{\xi}_i$

divide by S : $\frac{1}{S}\tilde{\chi}_S \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{S}\right), \quad \text{std dev } \frac{\sigma}{\sqrt{S}}$

Example same coin: $\tilde{\chi}_S$ = number of heads in S tosses, $\frac{1}{S}\tilde{\chi}_S$ = fraction of heads



bars: $\mathbb{P}(\frac{1}{S}\tilde{\chi}_S = \cdot)$ curve: $\mathcal{N}(\frac{1}{2}, \frac{1}{4S})$

$$S = 100: \quad \frac{\sigma}{\sqrt{S}} = \frac{1/2}{10} = 0.05, \quad \mathbb{P}(0.40 \leq \frac{1}{S}\tilde{\chi}_S \leq 0.60) \approx 0.95 \quad (\mu \pm 2\sigma/\sqrt{S})$$

S	1	100	10,000
$\sigma/\sqrt{S}$	0.5	0.05	0.005

10× more precise needs 100× more samples

$S = 1$: one decision, made once — nothing averages out; the spread stays σ

2. Duality theory

in this course, we focus on **convex** functions.

Primal Problem

$$\begin{array}{ll} \inf & f_0(x) \\ \text{s.t.} & x \in \mathbb{R}^N \\ & f_i(x) \leq 0 \quad \forall i \in [I] \end{array} \quad (P)$$

Dual Problem

$$\begin{array}{ll} \sup_{\theta} & \inf_{x \in \mathbb{R}^N} f_0(x) + \sum_{i \in [I]} \theta_i f_i(x) \\ \text{s.t.} & \theta \in \mathbb{R}^I \\ & \theta \geq 0 \end{array} \quad (D)$$

$\begin{array}{ll} \inf & f_0(x) \\ \text{s.t.} & x \in \mathbb{R}^N \\ & f_i(x) \leq 0 \quad \forall i \in [I] \end{array}$
$= \inf_{x \in \mathbb{R}^N} f_0(x) + \underbrace{\sup_{\theta \in \mathbb{R}_+^I} \sum_{i \in [I]} \theta_i f_i(x)}_{\leq 0}$

$$\begin{array}{ll} \text{if } f_i(x) \leq 0 \quad \forall i, & \text{then } = 0 \\ \text{if } \exists i : f_i(x) > 0, & \text{then } = +\infty \end{array}$$

$$\geq \sup_{\theta \in \mathbb{R}_+^I} \inf_{x \in \mathbb{R}^N} f_0(x) + \sum_{i \in [I]} \theta_i f_i(x)$$

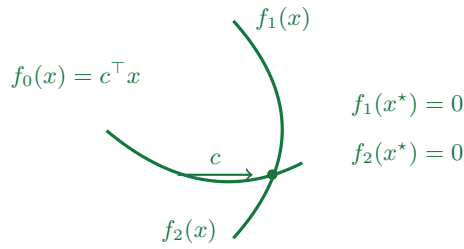
Properties:

— **weak duality** $\inf(P) \geq \sup(D)$ always holds.

— **strong duality:** $\inf(P) = \sup(D)$ holds

when the problem (P) is convex and the Slater's condition holds

$$\exists x \in \mathbb{R}^N, \text{ s.t. } f_i(x) < 0 \quad \forall i \in [I]$$



In this course, we assume that strong duality holds for convex problem!!

3. Example: linear program

Primal LP:

$$\begin{aligned} \inf \quad & c^\top x \\ \text{s.t.} \quad & x \in \mathbb{R}^N \\ & Ax \leq b \end{aligned} \quad \begin{aligned} f_0(x) &= c^\top x \\ f_i(x) &= a_i^\top x - b_i \quad \forall i \in [I] \end{aligned}$$

$$\begin{aligned} f_i(x) &\leq 0 \\ \iff a_i^\top x - b_i &\leq 0 \\ \iff a_i^\top x &\leq b_i \end{aligned}$$

Dual LP:

$$\begin{aligned} & \sup_{\theta \in \mathbb{R}_+^I} \inf_{x \in \mathbb{R}^N} c^\top x + \sum_{i \in [I]} \theta_i (a_i^\top x - b_i) \\ &= \sup_{\theta \in \mathbb{R}_+^I} \inf_{x \in \mathbb{R}^N} c^\top x + \sum_{i \in [I]} \theta_i a_i^\top x - \sum_{i \in [I]} \theta_i b_i \\ &= \sup_{\theta \in \mathbb{R}_+^I} - \sum_{i \in [I]} \theta_i b_i + \underbrace{\inf_{x \in \mathbb{R}^N} \left(c + \sum_{i \in [I]} \theta_i a_i \right)^\top x}_{= \begin{cases} 0 & \text{if } c + \sum_{i \in [I]} \theta_i a_i = 0 \\ -\infty & \text{if otherwise.} \end{cases}} \\ &= \sup_{\theta \in \mathbb{R}_+^I} -b^\top \theta \\ & \text{s.t.} \quad A^\top \theta = -c \end{aligned}$$

LP strong duality : if there exists $x \in \mathbb{R}^N : Ax \leq b$

$$\text{or } \theta \in \mathbb{R}^I : A^\top \theta = -c, \theta \geq 0$$

then strong duality holds

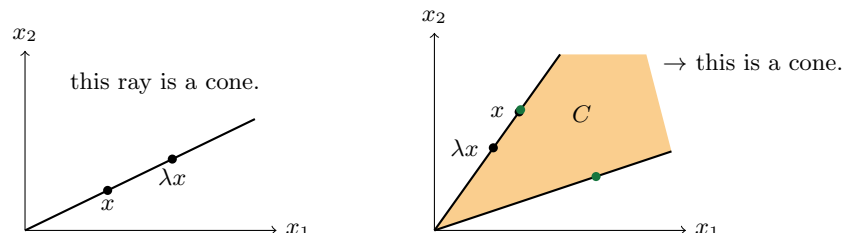
4. Conic programming

a conic program is a convex optimization problem.

— a **linear** objective function.

— a feasible set defines through the intersection of halfspaces $Ax \leq b$ or a proper cone.

Def (cone) a set C is a cone if $\forall x \in C$, we have $\lambda x \in C \quad \forall \lambda \geq 0$



Def (proper cone) $\mathbb{R}^N$ is a cone but not pointed.

C is a proper cone if it's **convex, closed, pointed**

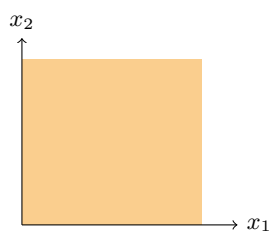
if $x, -x \in C$ then $x = 0$.

Generic conic program:

$$\begin{aligned} \inf \quad & c^\top x \\ \text{s.t.} \quad & x \in \mathbb{R}^N \\ & Ax \preceq_c b \end{aligned} \quad \Longleftrightarrow \quad b - Ax \in C$$

Example:

1) **Linear program:** $C = \mathbb{R}_+^K$



$$\begin{aligned} b - Ax &\in \mathbb{R}_+^K \\ \Longleftrightarrow b - Ax &\geq 0 \quad \Longleftrightarrow \quad Ax \leq b \end{aligned}$$

2) **Semidefinite program:** $C = \mathbb{S}_+^K$

$\mathbb{S}^K$: set of symmetric matrices in $\mathbb{R}^{K \times K}$ ($A^\top = A$)

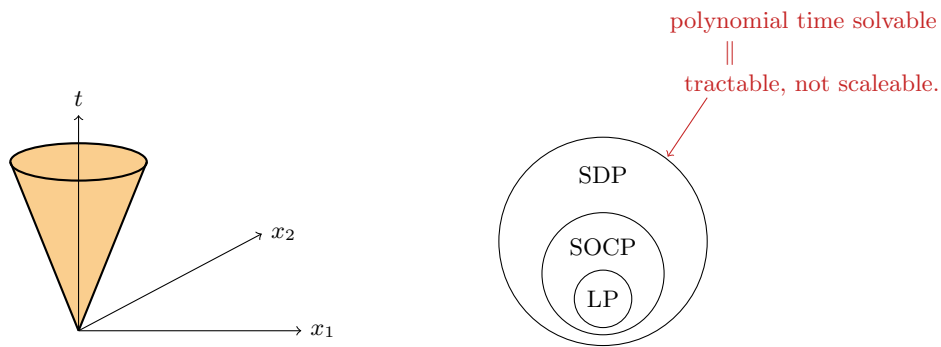
$\mathbb{S}_+^K$: set of positive semidefinite (symmetric) matrices in $\mathbb{R}^{K \times K}$

$A \in \mathbb{S}_+^K$ if $z^\top A z \geq 0 \quad \forall z \in \mathbb{R}^K$

Generic semidefinite program:

$$\begin{aligned} \inf \quad & c^\top x \\ \text{s.t.} \quad & x \in \mathbb{R}^N \\ & A_1 x_1 + A_2 x_2 + \dots + A_N x_N \preceq_{\mathbb{S}_+^K} B \end{aligned} \quad \rightarrow \text{usually ignored}$$

3) **Second-order cone:** $C = \{(x, t) \in \mathbb{R}^N \times \mathbb{R} : \|x\|_2 \leq t\}$ “ice-cream” cone



5. Conic programming duality

Def (*dual cone*)

$$C^* = \{y \in \mathbb{R}^K : \langle y, x \rangle \geq 0 \quad \forall x \in C\}$$

Self dual cones ($C = C^*$)

— $C = \mathbb{R}_+^K$

— $C = \mathbb{S}_+^K$

— $C = \text{second-order cone}$

Primal conic program:

$$\begin{aligned} \inf \quad & c^\top x \\ \text{s.t.} \quad & x \in \mathbb{R}^N \\ & Ax \preceq_c b \end{aligned}$$

Dual conic program:

$$\begin{aligned} \sup \quad & -\theta^\top b \\ \text{s.t.} \quad & \theta \in C^* \\ & A^\top \theta = -c \end{aligned}$$